

When Pretty Pictures Fail

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EASY EQUATIONS...

$$5x - 6 = 0$$

GRADE SCHOOL $\rightarrow$ GRAD SCHOOL...

$$x^5 - 5x + 6 = 0$$

$$5x - 6 = 0$$

$$\begin{aligned} 2x - 3y &= 1 \\ 5x - 6y &= 2 \end{aligned}$$

GRADE SCHOOL $\rightarrow$ GRAD SCHOOL...

How do you write
down solutions?

$$x^2 - 5x + 6 = 0$$



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$$\{2, 3\}$$

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$$x^{\boxed{3}} - 5x + 6 = 0$$

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GRADE SCHOOL $\rightarrow$ GRAD SCHOOL...

$$-\frac{\sqrt[3]{81+3\sqrt{354}}}{3} - \frac{5}{\sqrt[3]{81+3\sqrt{354}}} \text{ and two others...}$$

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GRADE SCHOOL $\rightarrow$ GRAD SCHOOL...



Scipione del Ferro
1465.02.06 – 1526.11.5

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and three others...

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Lodovico Ferraria
1522.02.02 – 1565.10.05

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GRADE SCHOOL $\rightarrow$ GRAD SCHOOL...

Complex Analysis

$$x^2 - 5x + 6 = 0$$



$$5x - 6 = 0$$

HYPERGEOMETRIC FUNCTIONS

No root of $x^5 - 5x + 6$ is expressible in terms of radicals [Niels Henrik Abel, 1824]...

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$$x = \sum_{i=0}^{\infty} \frac{(-1)^{\ell} (5\ell)!}{\ell! (4\ell+1)!} \cdot \frac{6^{4\ell+1}}{5^{5\ell+1}}.$$

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This is a **hypergeometric** series, like

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$$

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How do you solve
thousands of equations
in **thousands** of variables?

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Linear Algebra

$$2x - 3y = 1$$

$$5x - 6y = 2$$

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Linear Algebra

Complex Analysis

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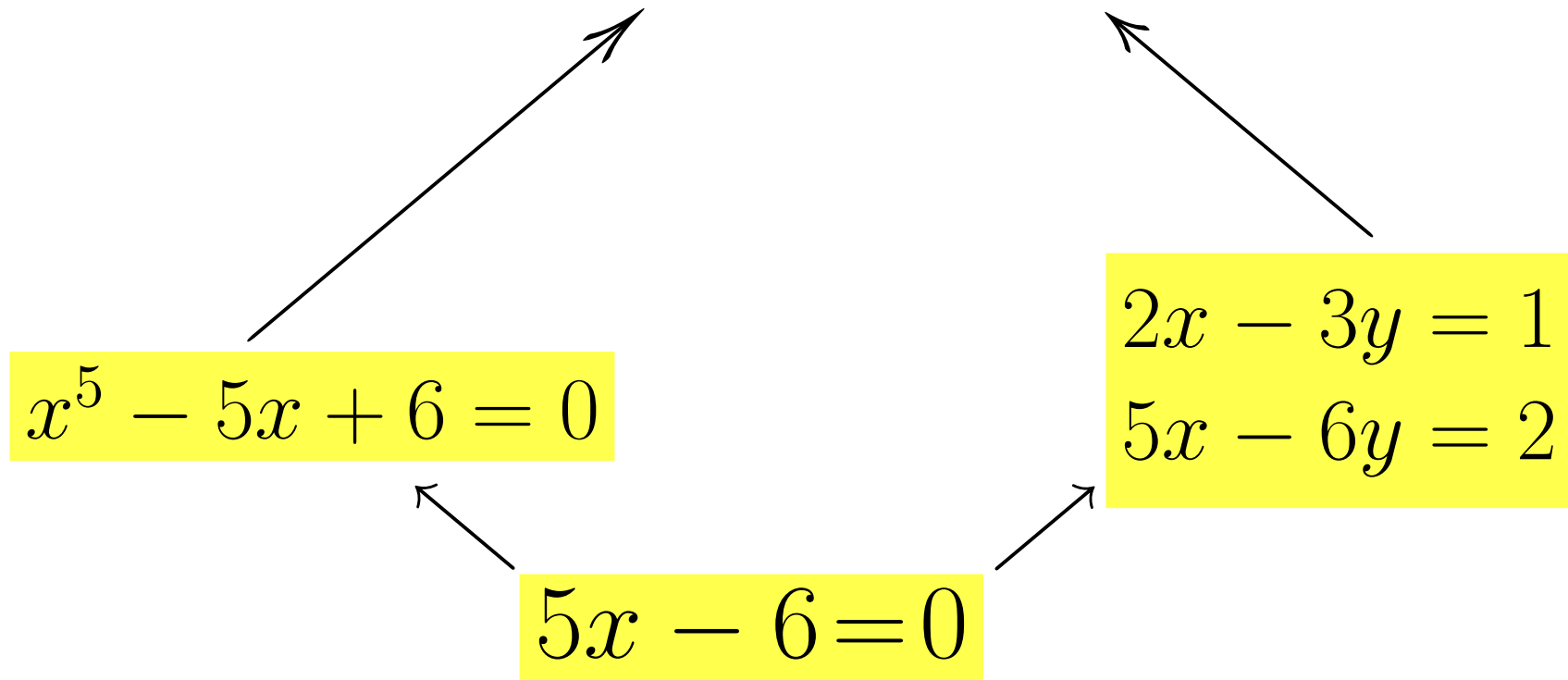

GRADE SCHOOL $\rightarrow$ GRAD SCHOOL...

Algebraic Geometry

$$x^5 - 5x + 6 = 0$$

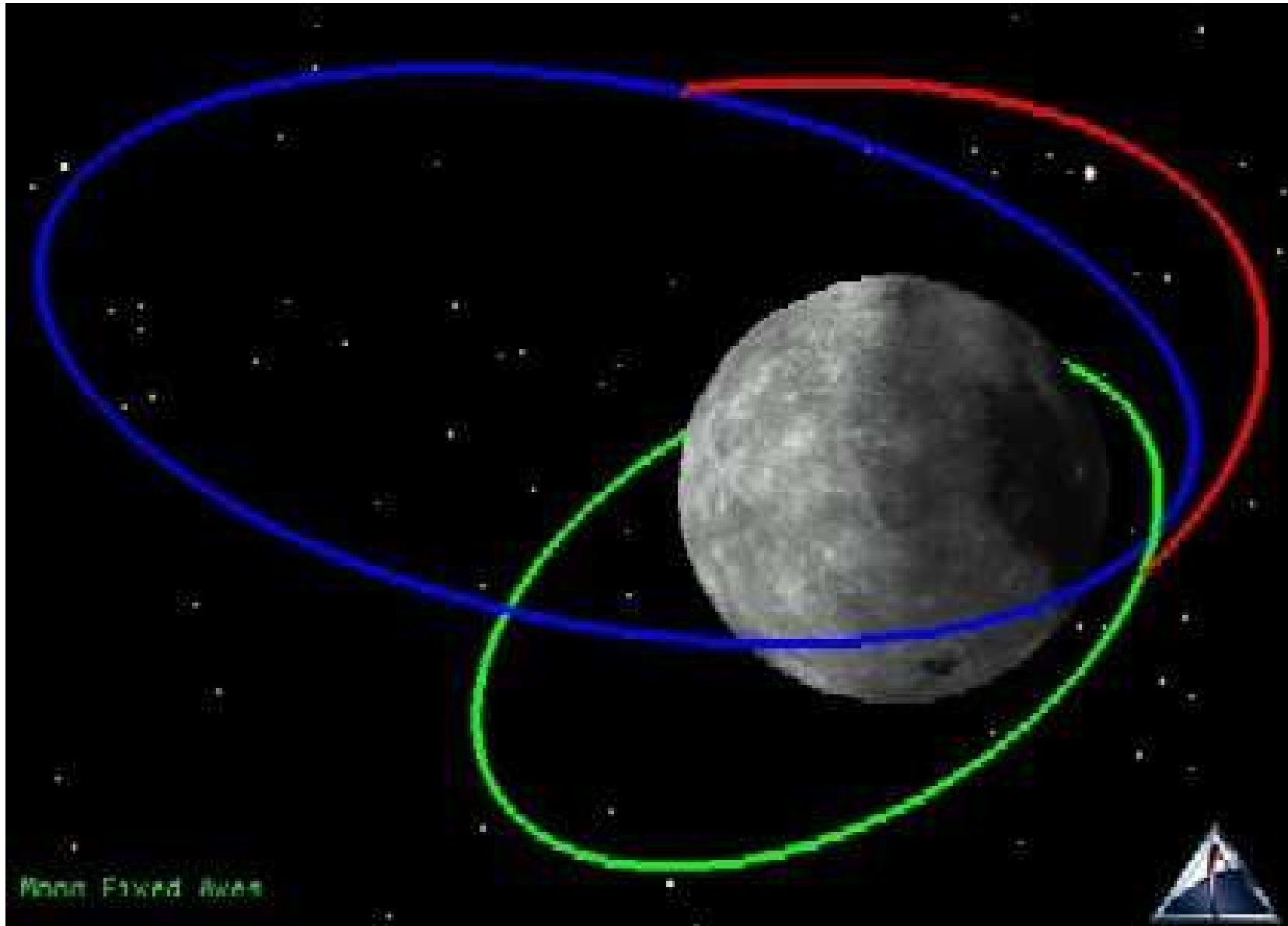
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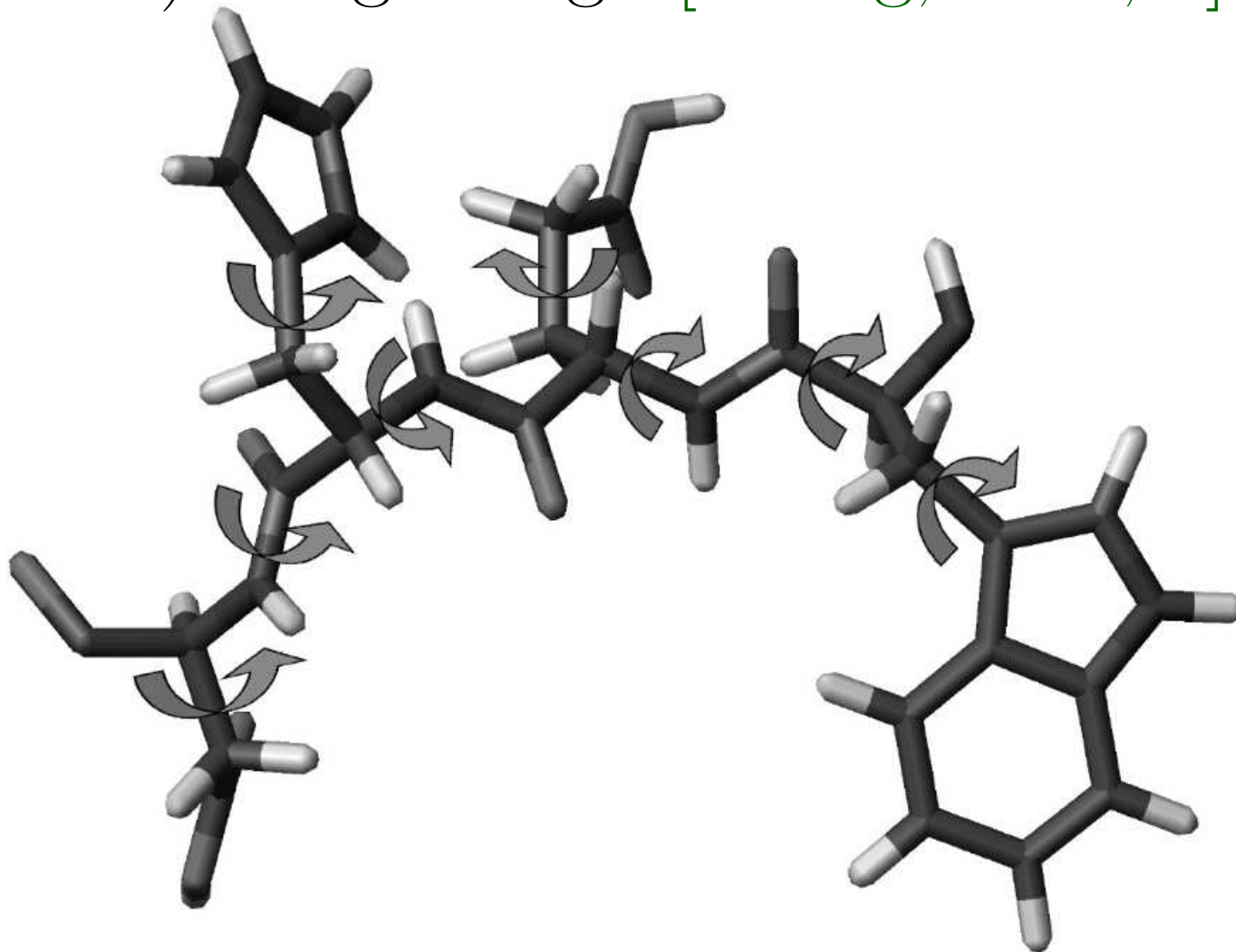
APPLICATION #1

Multi-Impulse Orbit Transfer [Mortari & Avendaño, 2008]...



APPLICATION #2

(Rational) Drug Design [Zhang, 2003;...]



$3n$ twist angles $\implies 3n$ equations in $3n$ unknowns...

APPLICATION #3

Phylogenetics

Point mutations in DNA sequences can be modelled probabilistically, leading to **hidden tree models** [Pachter-Sturmfels, 2004]. Such models then easily lead to **HUGE** polynomials systems, and the solutions are then probabilities that one organism ultimately evolved into another (within a fixed tree of organisms).

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In particular, we again care about **real** solutions...

APPLICATION #4



Probabilistic Modeling of Shock Response Spectrum Data

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Fitting the right probabilistic model relies on **algebraic statistics** [...; Pistone, Riccomagno, & Wynn, 2001; Drton, Sturmfels, Sullivan, 2009]...

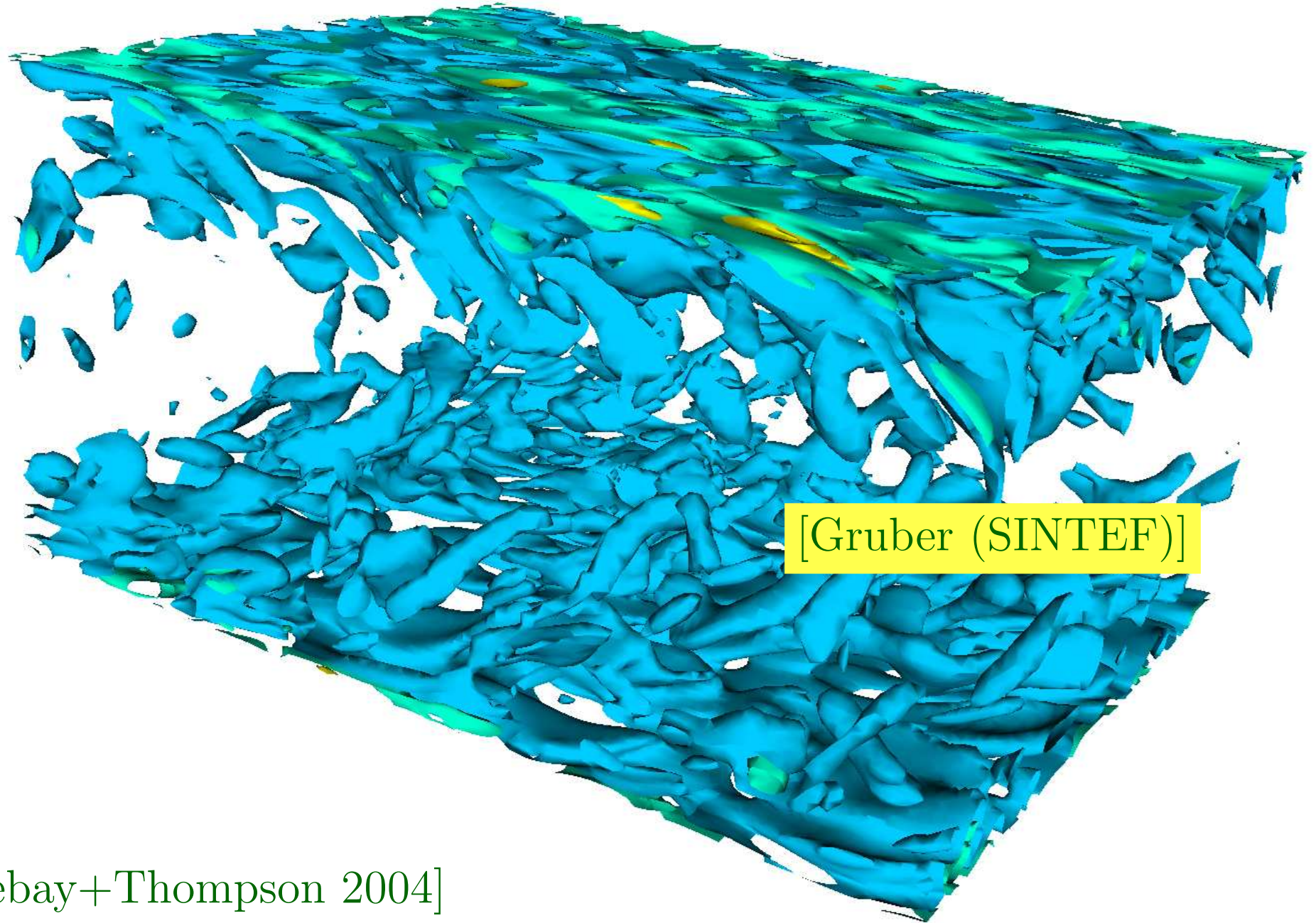
APPLICATION #4



Probabilistic Modeling of Shock Response Spectrum Data

Fitting the right probabilistic model relies on **algebraic statistics** [...; Pistone, Riccomagno, & Wynn, 2001; Drton, Sturmfels, Sullivant, 2009], and again reduces to finding **real** roots (usually in $[0, 1]$ or a **simplex**) of polynomial systems with **many** complex roots.

#5 FLOW FEATURE EXTRACTION



[Pebay+Thompson 2004]

THE RIGHT TOOL?

Suppose you want to find the exact number of positive roots of

$$x^5 - 5x + 6...$$

SOME HISTORY

≤1829: Jacques C. F. Sturm: The number of distinct roots of any real univariate polynomial in an interval **(a,b]** can be computed exactly via the number of sign alternations in two division-like sequences...



STURM SEQUENCES...

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...now count sign alternations in $(f_0(t), f_1(t), \dots, f_{57}(t))$ for $t=0$ and $t=+\infty$, and then subtract. (Should get **0** here.)

BIGGER?

Suppose you want to find the exact number of positive roots of

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$$\begin{aligned}f_0 &:= 1 - 2x^{196418} + x^{317811} \\f_1 &:= f'_0 = -392836x^{196417} + 317811x^{317810}\end{aligned}$$

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$$f_{58} := 0$$

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NOTATION

$$\text{Newt}(7 - 2.5x_1^2 + 27x_1^7x_2^5 + \sqrt{7}x_1^4x_2^4 + ex_2^7) = \dots$$

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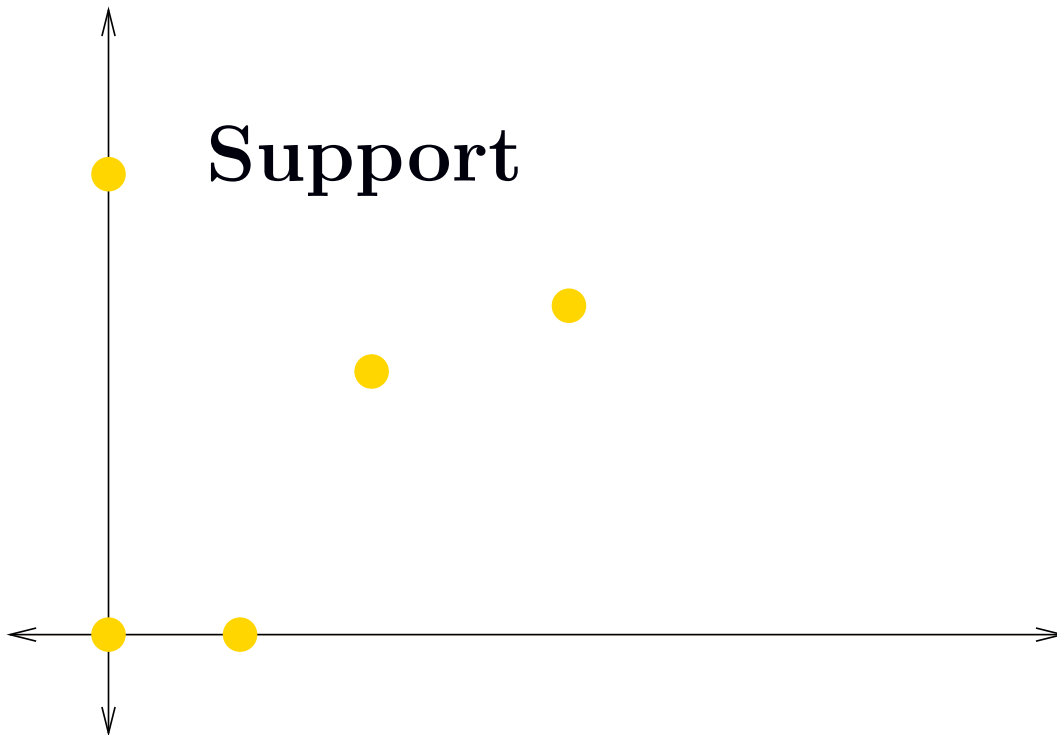
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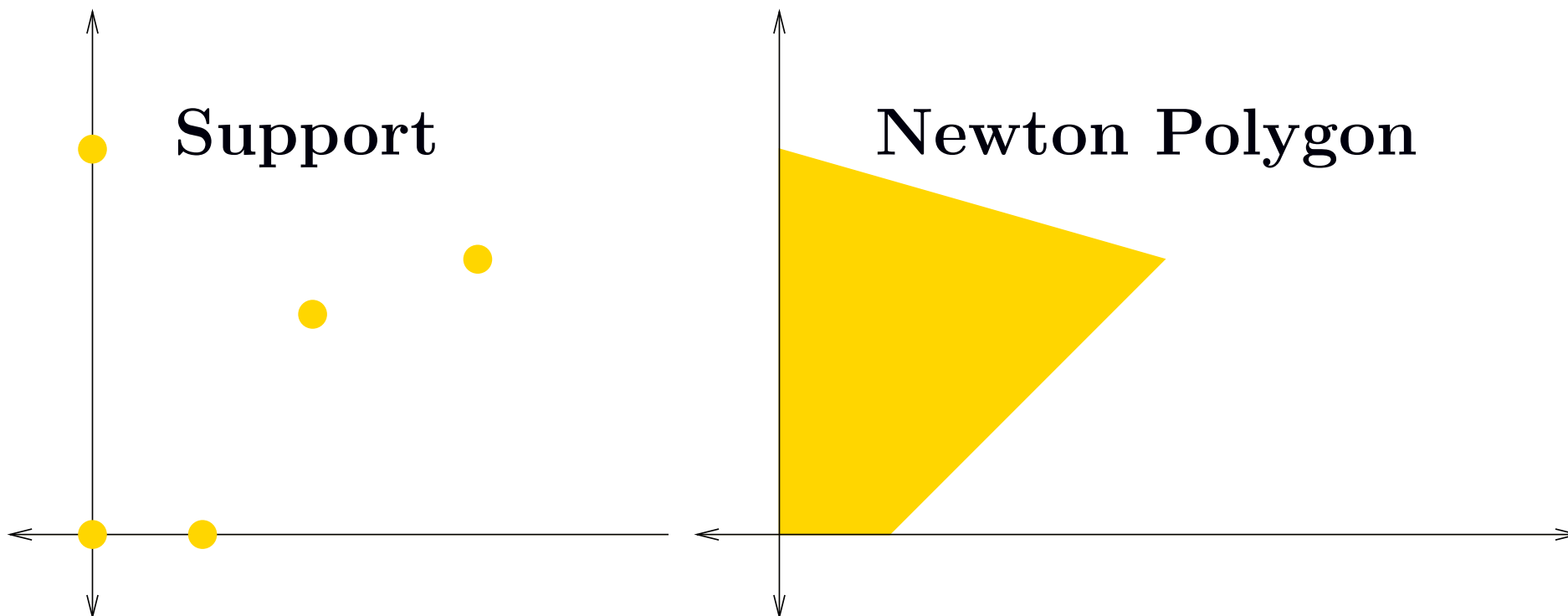
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POLYNOMIALS TO POLYHEDRA...

Consider $f(x, y) := 1 - x^2 + y - x^7y^5 + x^6y^7$. Then defining

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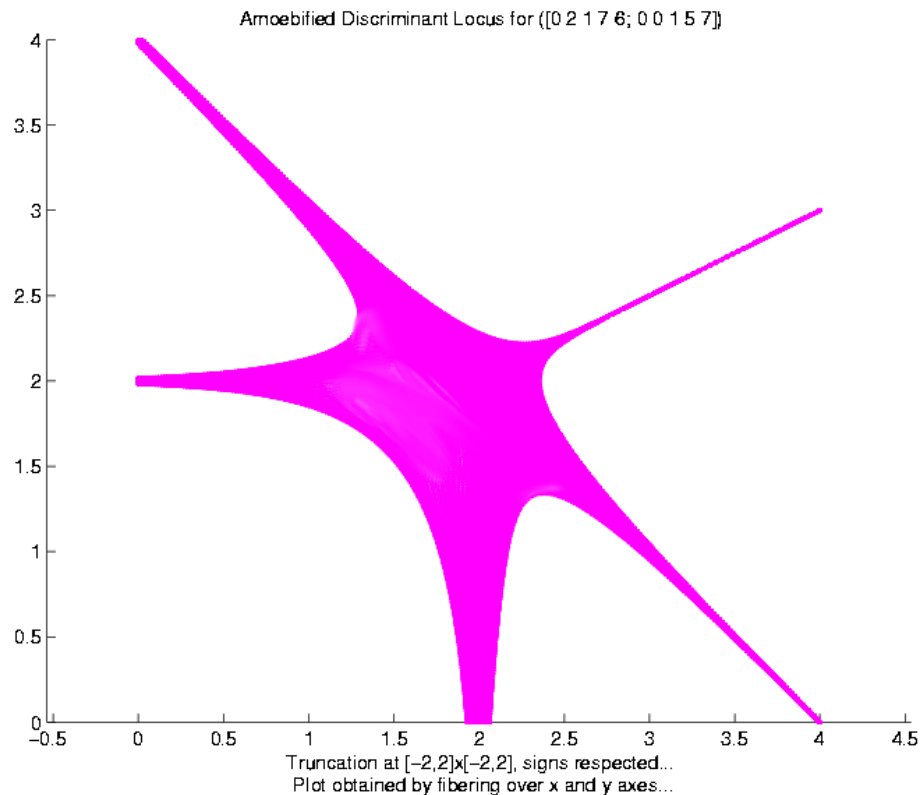
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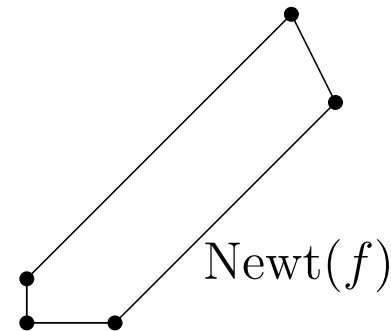
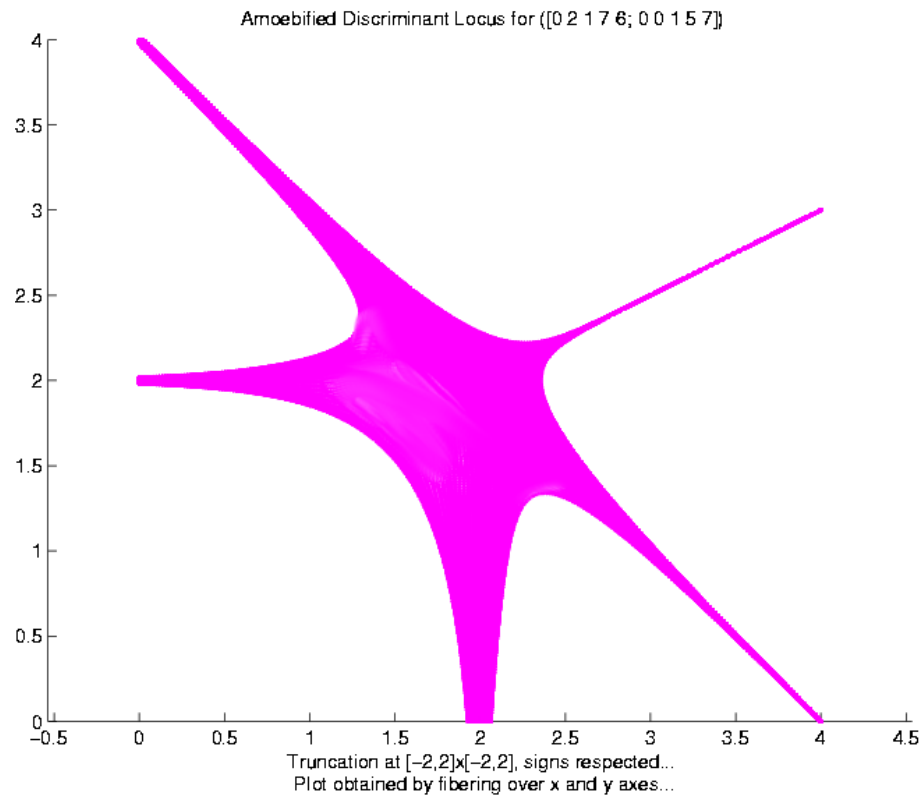


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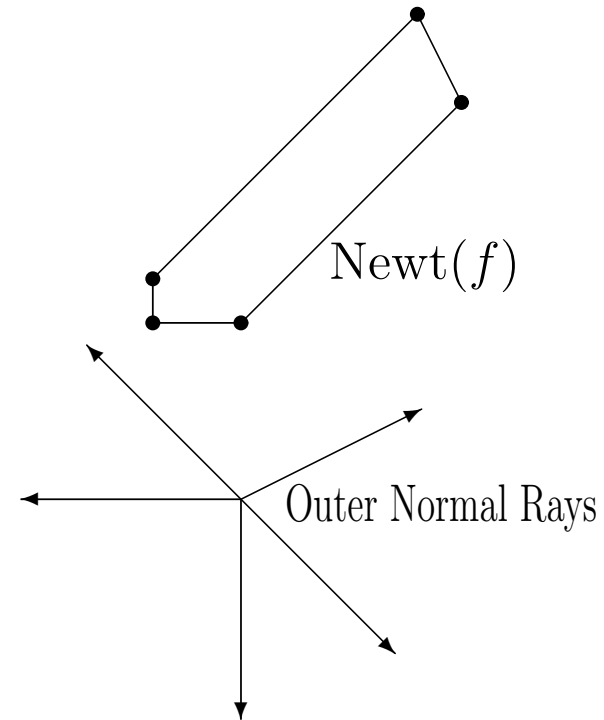
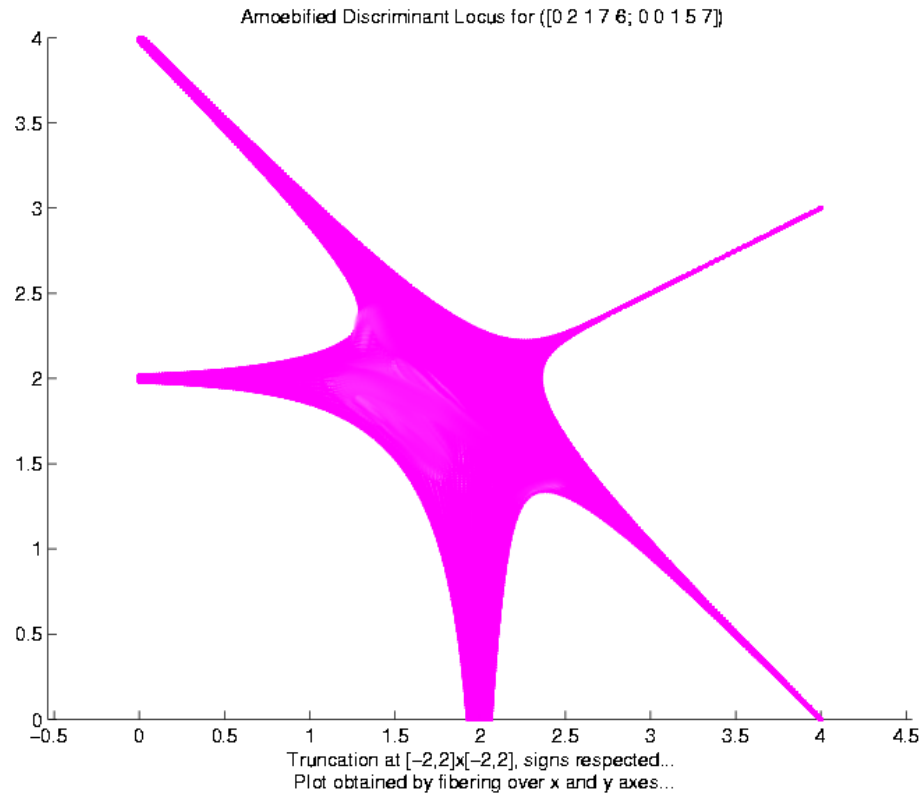


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So can we use combinatorics to **easily** predict where complex roots lie?

./movie



Thank you for listening!

Please see...

www.math.tamu.edu/~rojas

for on-line papers and further information.

This talk is downloadable from:

www.math.tamu.edu/~rojas/high.pdf